Figure 1

$$(2)\quad \{\{H_1..H_n\} \bullet \{A_1..A_n\}\} \longleftrightarrow \{E_1..E_n\}$$

$$(T1)\quad \{\{H_1..H_n\} \bullet \{A_1..A_5..A_n\}\} \longleftrightarrow \{E_1..E_5..E_n\}$$

$$(3)\quad \{\{H_1..H_n\} \bullet \{A_1..A_n\}\} \longleftrightarrow \{E_1..E_n\}$$

The various holistic notions found in the literature, such as web of belief, paradigm, theory and background knowledge are unpacked in this thesis and modeled as *hypertextual adjudicatory reasoning trails*. A theory T1 is not just seen as a hypothesis, some auxiliaries, and a body of evidence. Rather, any theory T1 is seen as a set of hypotheses, a set of auxiliaries, and a set of evidence. Each item within a set is itself seen as a node, the acceptance of which is hypertextually linked with another reasoning trail of hypotheses, auxiliaries, and set of evidence. For instance, in the above diagram, the auxiliary T1-A_5 is accepted by scientists at a particular time because they believe the trail (2) unproblematic. Similarly, they accept T1-E_5 because they believe the reasoning trail of (3) unproblematic. In the latter case, E_5 might be an observation made with a particular instrument. The reliability of the instrument is based upon accepting (3). In my thesis three points are made repeatedly against the relativist.

1. As a matter of pure logic our justificatory trails can go on forever; all hypertextual links are nonterminating and there is no foundational resting point for any proposition. However, this does not mean that scientists at a particular time do not have good ampliative reasons for accepting a particular node and its hyperlinked justification. For instance, given that H_1 in (2) is crucial to (2) which in turn supports A_5 in (T1), and given that H_1 could be examined or questioned at any time, does not mean that there are good reasons for doing so at any particular time. In Chapter 1 we will see that Feyerabend claims that scientists could have begun to question, via an examination of optical theory, the foundation for centuries of accepted naked-eye triangulations necessary for fixing planetary positions. That they did not do so, Feyerabend claims, shows that they were muddleheaded and thoughtless. My

point is that they were ampliatively "thoughtful," because there were no good reason for doubting the reliability of these observations. (Note also the independence from theoretical allegiance of the acceptance of these triangulations. See point 2 below.) Similarly, we have no good reason for doing another study on cigarette smoking and controlling for what people in a study did on their third birthdays, examining on that date what they ate, what environment they were in, and so on -- even though without studying such a variable, and innumerable others, there remains *mere conceivability* that we are wrong about the link between cigarette smoking and lung cancer.

2. The reasoning trails (2) and (3) can be and often are independent of T1. That is, their acceptance is often based on their own links and such acceptance may be involved in supporting a node on a rival to T1 as well. In other words, the myth of the framework has caused many to misplace the notion of and/or overemphasize the hegemonic force of theory ladeness. E_5 in T1 may be theory laden in terms of relying on (3), but (3) may not only be independent of T1 but also be used by (hypertextually linked to) a rival to T1. The acceptance of a telescopic observation may be linked with a reasoning trail, but that observation could be part of an evidence set of a rival theory. See Chalmers point against Feyerabend, Chapter 1.

3. When a particular node and its hypertextual justification are questioned, say E_5 and (3), an examination of the justification can be carried out independent of T1. For instance, Feyerabend argues correctly that during Galileo's time a rival to (3) existed (the celestial-terrestrial distinction and related matters), such that if it were true, it would have destroyed the supporting structure of (3) for E_5 (say, the observation of mountains on the moon). However, I argue in Chapter 1 that a resolution of this matter and an examination of the reliability of the celestial-terrestrial distinction could be cared out independent of whether one accepted T1 (Copernicanism) or not.

The captivating oversimplification of such notions as web of belief (Quine), paradigm (Kuhn), discourse recontextualizations (Rorty), are partly responsible for many postmodernists missing the many instances of piecemeal rational change in the history of science. I also argue that the messiness and contingency revealed by historical analyses are best captured by seeing scientists not only immersed in idiosyncratic contexts, but also attempting to make intelligent ampliative linkages modeled above. In this way an overall rational scaffolding is preserved, but an inevitable residue of disquiet and insecurity remain (allowing for robust disagreements), because of the nonterminating nature of all the links and because it is seldom clear what all the links should be at any given time. At no time in the history of science do we find a whole scientific community experiencing the non-argumentative comfort implied by Kuhn's notions of paradigm and normal science. Even during the monumentally successful Ptolemaic era we find the participants debating a whole host of issues -- the reality of epicycles, the appropriateness of the equant point, the unhappy marriage of Ptolemaic geometry and Aristotelian physics, the possibility of replacing Aristotelian physics with some sort of impetus theory, and so on.